Chapter 6

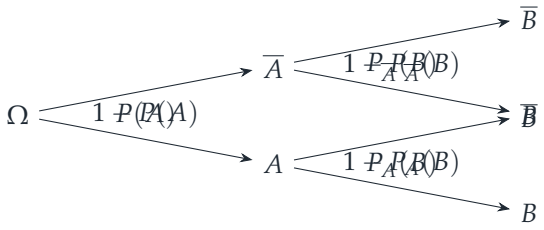
Conditional probabilities

Skills: model with a tree or a table, calculate, reason about independence, interpret

6.1 Reference notes

For two events A and B with $P(A) > 0$, the probability of B given A is

$$P_A(B) = P(B \mid A) = \frac{P(A \cap B)}{P(A)}.$$



Key ideas

In a weighted tree, the probabilities on the branches leaving the same node sum to 1. We multiply the probabilities along a path: $P(A \cap B) = P(A)P_A(B)$. We add the probabilities of disjoint paths leading to the required event.

If $A_1, \dots, A_n$ form a partition of the sample space, then

$$P(B) = P(A_1)P_{A_1}(B) + \cdots + P(A_n)P_{A_n}(B).$$

Two events A and B with non-zero probabilities are independent if $P(A \cap B) = P(A)P(B)$, or equivalently if $P_A(B) = P(B)$.

Common mistake

Do not confuse $P_A(B)$ and $P_B(A)$. The condition appears after the word "given" and is written as a subscript in the French notation.

Worked example

40% of customers buy online. Among them, 12% return their order; among the others, 5% return it. Then $P(R) = 0{,}40 \times 0{,}12 + 0{,}60 \times 0{,}05 = 0{,}078$. The overall return rate is so 7.8%.

Session 39 — Conditional probability

Conditional probabilities

09/03/2027 • B : 2 h – C : 1 h

Objectives of the session

· Define a conditional probability.

· Go from frequencies to probabilities.

· Interpret.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 20% of 140. . . . . .

2. What is the sum of the branches leaving a node?

3. Calculate $0.7 \times 0{,}4$. . . . . .

4. If $P(A) = 0.37$, calculate $P(\overline{A})$. . . . . .

5. Simplify $\dfrac{18}{30}$. . . . . .

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today. Before the activity, have students rephrase the key words: define a conditional probability and move from frequencies to probabilities.

Investigation

In a survey, 180 of 600 customers use a premium card, and 72 of these 180 customers buy every week. Calculate $P(P)$, $P(P \cap H)$ and $P_P(H)$, then interpret.

Written summary

For $P(A) > 0$, $P_A(B) = \dfrac{P(A \cap B)}{P(A)}$. This is the probability of B in the population restricted to A.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. We know $P(A) = 0{,}35$, $P_A(B) = 0.6$ and $P_{\overline{A}}(B) = 0.8$. Build the tree.

2. Calculate $P(A \cap B)$ then $P(B)$.

3. Calculate $P_B(A)$ and test independence.

References in the textbook

Extensions: Calao, lesson pp. 18-21; exercises p. 24, nos. 1-3. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 138, Exercise 5

A company buys electronic chips from three suppliers F_1, F_2 and F_3 : 30% from F_1, 20% from F_2. Some chips are defective: 5% of the chips coming from F_1, 3% of those from F_2 and 10% of the chips from F_3.
A chip is chosen at random. Consider the events F_i : "The chip comes from supplier i " and D : « The chip is defective".

1. Complete the weighted tree from the textbook.

2. Determine the probability that the chosen chip is defective.

3. Given that the chip is defective, determine the probability that it comes from supplier F_3.

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 315 and the rate or common ratio with 5. Write four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 40 — Weighted trees

Conditional probabilities

11/03/2027 • B : 1 h – C : 2 h

Objectives of the session

· Build and complete a tree.

· Read a conditional probability.

· Check the sums.

Fluency · 6 to 8 minutes, without a calculator

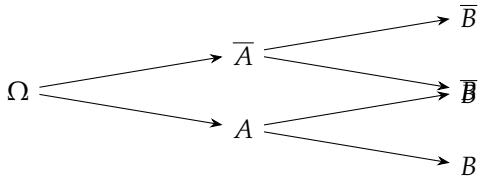
1. Write the multiplier for a 25% increase
2. Calculate $0.8 \times 0{,}4$. .....
3. If $P(A) = 0.37$, calculate $P(\overline{A})$. .....
4. Simplify $\dfrac{18}{30}$. .....
5. What is the sum of the branches leaving a node?

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: construct and complete a tree and read a conditional probability.

Investigation

35% of orders are express. Among them, 4% are late; among standard orders, 9% are late. Construct and complete the tree.



Written summary

In a tree, the branches leaving a node sum to 1. The second-level probabilities are conditional.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. We know $P(A) = 0{,}40$, $P_A(B) = 0.2$ and $P_{\overline{A}}(B) = 0.4$. Build the tree.
2. Calculate $P(A \cap B)$ then $P(B)$.

3. Calculate $P_B(A)$ and test independence.

References in the textbook

Extensions: Calao, lesson pp. 18-21; exercises p. 24, nos. 1-3. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 140, Exercise 15

1. Complete the tree from the textbook, given that $P(A) = \frac{3}{5}$, $P_A(B) = \frac{7}{10}$, $P_A(D) = \frac{1}{5}$, $P_{\overline{A}}(B) = \frac{1}{3}$ and $P_{\overline{A}}(C) = \frac{2}{9}$.
2. Determine $P(A \cap B)$, $P(A \cap D)$ and $P(C)$.

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 320 and the rate or common ratio with 6. Write four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Session 41 — Product of the probabilities along a path

Conditional probabilities

16/03/2027 • B : 2 h – C : 1 h

Objectives of the session

· Calculate an intersection.

· Combine paths.

· Translate an event.

Fluency · 6 to 8 minutes, without a calculator

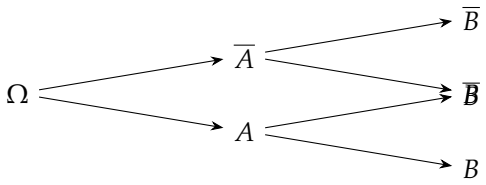
1. Simplify 3 $x + 9 - x + 2$. .....
2. If $P(A) = 0.37$, calculate $P(\overline{A})$. .....
3. Simplify $\dfrac{18}{30}$. .....
4. What is the sum of the branches leaving a node?
5. Calculate $0.9 \times 0{,}4$. .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: calculate an intersection and combine paths.

Investigation

With the tree from session 40, calculate the probability that an order is express and late, then the total probability of a delay. Interpret both results.



Written summary

We multiply along a path: $P(A \cap B) = P(A)P_A(B)$. We add the probabilities of disjoint paths which make up the event.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. We know $P(A) = 0{,}45$, $P_A(B) = 0.3$ and $P_{\overline{A}}(B) = 0.5$. Build the tree.
2. Calculate $P(A \cap B)$ then $P(B)$.

3. Calculate $P_B(A)$ and test independence.

References in the textbook

Extensions: Calao, p. 140, nos. 14 to 18. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the digital resources of the textbook.

Calao textbook · p. 140, Exercise 18

75% of a restaurant's customers are families. Among them, 17% pay their bill with holiday vouchers, 76% pay their bill by bank card and the others pay in cash.
Of the other customers, only 5% use holiday vouchers, 80% a bank card and the others cash.
A table and its customer(s) are chosen at random. Let F : "The customers are a family", C : "The customers pay with holiday vouchers", B : "The customers pay by bank card" and L : « The customers pay in cash".

1. Complete the weighted tree from the textbook.
2. Describe in a sentence the event $\overline{F} \cap B$ then calculate its probability.
3. Calculate the probability that the chosen customers pay with holiday vouchers.
4. The selected customers paid with holiday vouchers. Calculate the probability that these customers are a family.

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 325 and the rate or common ratio with 7. Write four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 42 — Independence

Conditional probabilities

18/03/2027 • B : 1 h – C : 2 h

Objectives of the session

· Test independence.

· Compare probabilities.

· Justify a conclusion.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 5% of 60.
2. Simplify $\frac{18}{30}$.
3. What is the sum of the branches leaving a node?
4. Calculate $0.3 \times 0{,}4$.
5. If $P(A) = 0.37$, calculate $P(\overline{A})$.

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: test independence and compare probabilities.

Investigation

In a population, $P(A) = 0{,}4$, $P(B) = 0{,}3$ and $P(A \cap B) = 0{,}12$. Test the independence of two ways. Then modify $P(A \cap B)$ to 0.15 and start again.

Written summary

A and B are independent if $P(A \cap B) = P(A)P(B)$, equivalently if $P_A(B) = P(B)$ when $P(A) > 0$.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. We know $P(A) = 0{,}20$, $P_A(B) = 0.4$ and $P_{\overline{A}}(B) = 0.6$. Build the tree.
2. Calculate $P(A \cap B)$ then $P(B)$.
3. Calculate $P_B(A)$ and test independence.

References in the textbook

Extensions: Calao, lessons pp. 134-137; exercises p. 144, nos. 33 to 35. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find

the figures and the textbook's digital resources.

Calao textbook · p. 139, Exercise 13

On leaving a factory, watches may have a defect (denoted D_1) on the strap and a defect (denoted D_2) on the dial. A watch is called defective if it has at least one of the two defects. A watch is selected at random from one day's production.
Let A the event: "The watch has the defect D_1" and B the event: "The watch has the defect D_2". The probabilities of these events are known: $P(A) = 0{,}02$ and $P(B) = 0{,}01$. We assume that these two events are independent.

1. Determine the probability that the watch has both defects.
2. Determine the probability that the watch has no defect.

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 330 and the rate or common ratio with 2. Write four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Session 43 — Partition and the law of total probability

Conditional probabilities

23/03/2027 • B : 2 h – C : 1 h

Objectives of the session

· Recognise a partition.

· Apply the law of total probability.

· Organise the calculation.

Fluency · 6 to 8 minutes, without a calculator

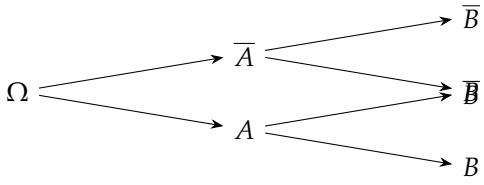
1. Write the multiplier for a 10% increase
2. What is the sum of the branches leaving a node?
3. Calculate $0.4 \times 0{,}4$. .....
4. If $P(A) = 0.37$, calculate $P(\overline{A})$. .....
5. Simplify $\dfrac{18}{30}$. .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: recognise a partition and apply the law of total probability.

Investigation

A factory has three lines A, B, C producing 20%, 50%, 30% of the items respectively.
Their defect rates are 1%, 2%, 4%. Calculate the overall defect rate.



Written summary

If (A_i) is a partition, $P(B) = \sum_i P(A_i)P_{A_i}(B)$. Each term corresponds to a path leading to B.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. We know $P(A) = 0{,}25$, $P_A(B) = 0.5$ and $P_{\overline{A}}(B) = 0.7$. Build the tree.

2. Calculate $P(A \cap B)$ then $P(B)$.

3. Calculate $P_B(A)$ and test independence.

References in the textbook

Extensions: Calao, p. 138, nos. 1 to 5, then p. 142, nos. 22 to 25. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 139, Exercise 7

In the early 2000s, the Internet was not as popular, and a survey of interest in the Internet among a sample of 2,000 people gave the following results.

Interested in the internet?	Yes	No	Total
Under 30	560	...	700
Aged 30 to 60	...	...	...
Over 60	...	510	600
Total	800	...	2 000

1. Complete this table.

2. A respondent is chosen at random. Let A_1 : "The person is under 30 », A_2 : "The person is between 30 and 60 years old", A_3 : "The person is over 60" and I : "The person is interested in the Internet". Using the table, complete the probability trees in the textbook.

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 335 and the rate or common ratio with 3. Write four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 44 — Problems in context

Conditional probabilities

25/03/2027 • B : 1 h – C : 2 h

Objectives of the session

· Choose a tree or a table.

· Solve a problem in context.

· Interpret.

Fluency · 6 to 8 minutes, without a calculator

1. Simplify $3\ \ x+5-x+2$. .....
2. Calculate $0.5\ \times 0{,}4$. .....
3. If $\ P(A)=0.37$, calculate $P(\overline{A})$. .....
4. Simplify $\ \dfrac{18}{30}$. .....
5. What is the sum of the branches leaving a node?

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: choose a tree or a table and solve a problem in context.

Investigation

Turn a two-way table of customers/subscription/cancellation into a tree, then answer a question both ways: probability of cancelling given the subscription; probability of having the subscription given cancellation.

Written summary

Choosing between a tree and a table: the tree highlights the order of conditioning; the table is well suited to frequencies and reverse readings.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. We know $P(A)=0{,}30$, $P_A(B)=0.6$ and $P_{\overline{A}}(B)=0.8$. Build the tree.
2. Calculate $\ P(A\cap B)$ then $\ P(B)$.
3. Calculate $\ P_B(A)$ and test independence.

References in the textbook

Extensions: Calao, p. 142, nos. 22 to 25, then p. 144, nos. 30 to 32. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 142, Exercise 24

A town council carries out a survey to find out the habits of its residents regarding waste management. 80% of respondents say they care about sustainable development, but among them only 75% sort their waste. 70% of the people surveyed say they sort their waste.
One of the people surveyed is chosen at random. Consider the events D : "The person surveyed cares about sustainable development" and T : "The person surveyed sorts their waste ».

1. Determine $P_D(T)$.
2. Determine $P_T(D)$.

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 340 and the rate or common ratio with 4. Write four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Session 45 — Mixed practice

Conditional probabilities

30/03/2027 • B : 2 h – C : 1 h

Objectives of the session

· Calculate an inverse probability.

· Link total probability and conditioning.

· Check the result.

Fluency · 6 to 8 minutes, without a calculator

1. Calculate 20 per cent of 120 . . . . . .

2. If $P(A) = 0.37$, calculate $P(\overline{A})$. . . . . .

3. Simplify $\dfrac{18}{30}$. . . . . .

4. What is the sum of the branches leaving a node?

5. Calculate $0.6 \times 0{,}4$. . . . . .

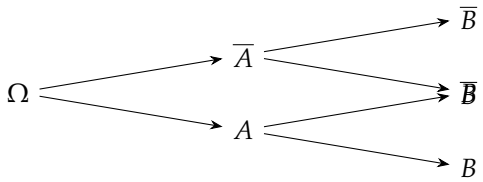
Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.

Before the activity, have students rephrase the key words: calculate a reverse conditional probability and combine total probability and conditioning.

Investigation

In the situation of session 43, given that an item is defective, calculate the probability that it comes from line C. Explain why this result is not 30%.



Written summary

For a reverse conditional probability, first calculate the total probability of the conditioning event, then use the definition of conditional probability.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. We know $P(A) = 0{,}35$, $P_A(B) = 0.2$ and $P_{\overline{A}}(B) = 0.4$. Build the tree.

2. Calculate $P(A \cap B)$ then $P(B)$.

3. Calculate $P_B(A)$ and test independence.

References in the textbook

Extensions: Calao, pp. 144-146. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and the digital resources of the textbook.

Calao textbook · p. 143, Exercise 26

In this exercise, results are to be given as fractions in lowest terms. A game consists in rolling a fair six-sided die then drawing a ball from an urn. If a 6 is rolled on the die, a ball is drawn from urn A; if an odd number is obtained, a ball is drawn from urn B; in the other cases, a ball is drawn from urn C.
Urn A contains 1 red ball and 5 green balls; urn B contains 2 red balls and 8 green balls; urn C contains 3 red balls and 11 green balls.
Consider the following events: I : "Roll an odd number"; S : "Roll a 6"; A : « Roll a 2 or a 4"; R : "The ball drawn is red"; V : "The ball drawn is green".

1. Draw a weighted tree for this situation.
2. Determine the probability of drawing a red ball.
3. Determine the probability that the ball came from urn A given that it is red.

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 345 and the rate or common ratio with 5. Write four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 46 — Review and assessment

Conditional probabilities

01/04/2027 • B : 1 h – C : 2 h

Objectives of the session

· Use all the methods.

· Write up using notation.

· Fix errors.

Fluency · 6 to 8 minutes, without a calculator

1. Write the multiplier for a 25% increase
2. Simplify $\frac{18}{30}$. .....
3. What is the sum of the branches leaving a node?
4. Calculate $0.7 \times 0{,}4$. .....
5. If $P(A) = 0.37$, calculate $P(\overline{A})$. .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today. Before the activity, have students rephrase the key words: use all the methods and write with the notation.

Investigation

Do a circuit of four stations: complete a tree; calculate an intersection; apply the law of total probability; test independence. Each group leaves a written justification for the next group.

Written summary

Always name the events, complete all branches, write the formula before calculating and interpret the final probability.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. We know $P(A) = 0{,}40$, $P_A(B) = 0.3$ and $P_{\overline{A}}(B) = 0.5$. Build the tree.
2. Calculate $P(A \cap B)$ then $P(B)$.
3. Calculate $P_B(A)$ and test independence.

References in the textbook

Extensions: Calao, pp. 122-123. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and the digital resources of the textbook.

Calao textbook · p. 143, Exercise 29

To hire its managers, a company uses a recruitment agency. The agency carries out an initial selection based on applications. 40% of the applications received are approved and passed on to the company. The selected candidates have a first interview, after which 70% of them are selected. These are invited to a final interview with the human resources director, who hires 25% of the candidates interviewed.
A candidate's file is randomly selected and the following events are considered: D : "The candidate is shortlisted on their application"; E_1 : "The candidate is selected after the first interview"; E_2 : "The candidate is recruited."

1. Complete the weighted tree from the textbook. Let F the event: "The candidate is not recruited ". Prove that the probability of the event F is equal to 0.93.

2. Five friends apply for a management job in this company. Their applications are reviewed independently of one another. Find the probability that at least one of the five friends is recruited.

3. What is the minimum number of applications the recruitment agency must process so that the probability of hiring at least one candidate is greater than 0.999?

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 350 and the rate or common ratio with 6. Write four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.